

Novel machine learning framework for thermal conductivity prediction by crystal graph convolution embedded ensemble

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Abstract

With the advent of the big data era, artificial intelligence technology has penetrated and deeply affected our daily life. In addition, data-based machine learning algorithms have been applied to physics, chemistry, material science, and other basic science fields. However, the scarcity of data sets is known as the main obstacle to its development. Mining effective information from the limited data samples and building an appropriate machine learning algorithms framework are the major breakthroughs. For solid materials, the intrinsic properties are closely related to their atomic composition and relative positions, namely crystal structures. Here, inspired by the emerging of graph convolution neural network and material crystal graph, we proposed an integrated algorithms framework embedded crystal graph to train and predict the lattice thermal conductivities of crystal materials. This machine learning algorithms framework showed superior learning and generalization ability. In addition, not only in predicting thermal conductivities, but our framework also has great performance in predicting other phonon or electron-related properties. This strategy provided a new approach in the design of machine learning framework, which indicated the great potential for the application of machine learning in material science.

KEYWORDS

crystal graph, ensemble algorithm, machine learning framework, thermal conductivity

1 | INTRODUCTION

Artificial intelligence (AI) technology has been widely used in our daily life. As the fourth paradigm of science (behind empirical, analytical, and simulation methods, proposed by Jim Gray in 2007), the data-intensive scientific method has aroused deep interests in material science, which is promising in predicting material

properties and designing new functional materials. Recently, machine learning algorithms had several preliminary explorations in evaluating material electronics, thermodynamics, magnetism, energy conversion, and chemical reaction properties.^{1–12} According to these, machine learning algorithms show the advantages of high-speed and low-consuming, compared with the traditional computational methods based on basic physical

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principles. Some machine algorithms have made considerable achievements, the lack of data sets and the scarcity of appropriate algorithms are still the key problems that restrict the development.¹⁴

Harvard Clean Energy Project,¹⁵ Materials Project,¹⁶ AFLOW,¹⁷ QM9,¹⁸ Open Quantum Materials Database (OQMD),¹⁹ Quantum MOF (QMOF)²⁰ and other database projects provide high-throughput computing frameworks and a large number of theoretical calculation samples. Besides, some databases have been engaged in collecting experimental data for a long time, such as Crystal Structure Database (CSD),²¹ Inorganic Crystal Structure Database (ICSD),²² and so on. However, limited by capacity and diversity of material properties, more adequate and comprehensive databases are still appealing. In terms of algorithms, the deficiency of data samples also hinders the direct application of some complex algorithms, for instance, the artificial neural network usually requires huge amounts of training data to learn from.²³ At present, the commonly used training algorithms are simple kernel-based or tree-based methods, such as Gaussian Process (GP), Support Vector Machine (SVM), Kernel ridge regression (KRR), decision trees (DT), and so on.^{10,11,13,24,25} Avoiding the limitations of single learner, constructing new algorithms frameworks are of great significance to predict material properties more accurately.

In this study, we focus on the thermal transport properties of crystal materials, where the lattice thermal conductivity (TC) is an important material intrinsic property that received widespread attention. With the increase of device integration, reasonable selecting and collocating materials with specific thermal transport properties can effectively control the working temperature, preventing device damage due to excessive Joule heat. On the other hand, the waste heat also causes serious energy losses. Thermoelectricity is an environmentally friendly energy conversion method, while excellent thermoelectric materials usually require lower lattice TC to ensure a large temperature gradient. Hence, understanding the thermal transport properties has the instructive significance for designing appropriate thermal functional materials. However, the first-principles calculation of the lattice TC is very time-consuming. Especially with the increasing of the material complexity, the calculation consumption would exhibit an explosive growth. Machine learning methods could be a good alternative in the prediction of thermal transport properties. Finding a more universal material feature extraction and training algorithms framework is urgent to be solved.

As is known, lattice thermal conductivities of crystal materials are mainly determined by the element composition and stacking structure, especially the atomic mass and the interaction between atoms. Inspired by the

recently proposed graph convolution network (GCN) and crystal graph of materials (CG),²⁶ in this study, we proposed a novel crystal graph convolution embedded ensemble machine learning algorithms framework and trained it to predict the TC. In this algorithms framework, the only input feature was the crystal structure. Crystal GCN can extract feature vectors from not only the atomic information but also their surrounding environments (chemical bonds or other hidden weak nonbond interactions). In addition, the boosting ensemble algorithm was used to overcome the limitations of one single weak learner.²⁷ Also, to improve the efficiency and prevent overfitting, we adopted the more ultimate extreme gradient boosting method (Xgboost),²⁸ proposed by Chen in 2017. The trained model has a higher accuracy and strong generalization ability in TC prediction. Moreover, this framework shows satisfactory metric scores in learning and predicting other phonon and electronic properties. Our well-performed algorithms framework shows great potential for the application of machine learning in material science.

2 | METHODS

2.1 | Data set

The data set is obtained through the computational materials databases constructed from high-throughput ab initio calculations using the AFLOW framework.^{17,29} The lattice thermal conductivities and other thermodynamic properties, such as the heat capacities and the Grüneisen parameters, are calculated according to the Automatic Gibbs Library (AGL) with quasiharmonic Debye–Grüneisen model approximation.^{30,31} Under this framework, thermal conductivities at the acoustic Debye temperature can be expressed using the Leibfried–Schlomann equation as,

$$\kappa_1(\theta_a) = \frac{0.849 \times 3\sqrt{4}}{20\pi^3(1 - 0.514\gamma^{-1} + 0.228\gamma^{-2})} \times \left(\frac{k_B \theta_a}{\hbar} \right)^2 \frac{k_B m V^{1/3}}{\hbar \gamma^2} \quad (1)$$

where V is the volume of the unit cell; m is the average atomic mass; θ_a is referred to the acoustic Debye temperature, can be calculated from the traditional Debye temperature θ_D . The thermal conductivities at T can be written as,

$$\kappa_1(T) = \kappa_1(\theta_a) \frac{\theta_a}{T} \quad (2)$$

We attached more than 5000 (5658) solid materials to collect their crystal structures and thermodynamic properties. These materials cover all 7 crystal systems and 128 space groups, as shown in Figure 1A. The cubic crystal systems structure materials account for the largest proportion, and the data samples account for 45% of the whole data set. The second is the hexagonal crystal system, accounting for about 20%. This can be ascribed to most of the high-performance functional materials are in high symmetry. This data set also covers samples with low symmetry, and the percentage of monoclinic, rhombohedral, and triclinic systems is about 1%. At the same time, according to their various electronic structure characteristics, this data set includes not only metallic conductors, but also semiconductors of direct or indirect bandgaps, as well as insulators with larger bandgaps. Samples in the data set contain most of the elements in the periodic table. The number of atoms in one unit cell is mostly 1–10, and individual materials contain dozens of atoms. The density distribution of these materials is also shown in Figure 1B. They are mainly located at densities below 10 g/cm³. Most importantly, the theoretical lattice conductivity materials with TC less than 1 W/(m K) and a few high TC materials, whose TC are more than 100 W/(m K). The lattice thermal conductivities of most samples spans between 1 and 50 W/(m K).

2.2 | Algorithms framework

In this study, we developed a brand-new crystal GCN embedded ensemble algorithms framework. According to Occam's razor principle, entities should not be multiplied unnecessarily, the only input feature in this framework is the crystal structures. The crystal structures can provide much useful information such as the atom species/numbers, as well as the bonding relationships between all the atoms. With crystal structures and target properties, the expected machine learning model can be realized by supervised training. The workflow of the whole algorithms is shown in Figure 1D.

First, we extracted the basic information from crystal structures (in *cif* or other structure file format of periodic theoretical calculation programs) and combined the atoms and bonding characteristics to form the crystal graph structure of each material. Different from graphs or pictures in our daily life, material crystal graphs concluded nodes and multiple edges. Each node encoded the property of one atom, which can be presented by a feature vector v_i . And edges represent the interaction features of the connecting atoms i and j . The edge feature vector, $u_{(i,j)k}$, encoding the k th interaction (bond or no-bond) between nodes (atom) i and j .

Then, the graph convolution neural network is used to extract the material feature vectors containing the basic atom features and their hidden bonding information.²⁶

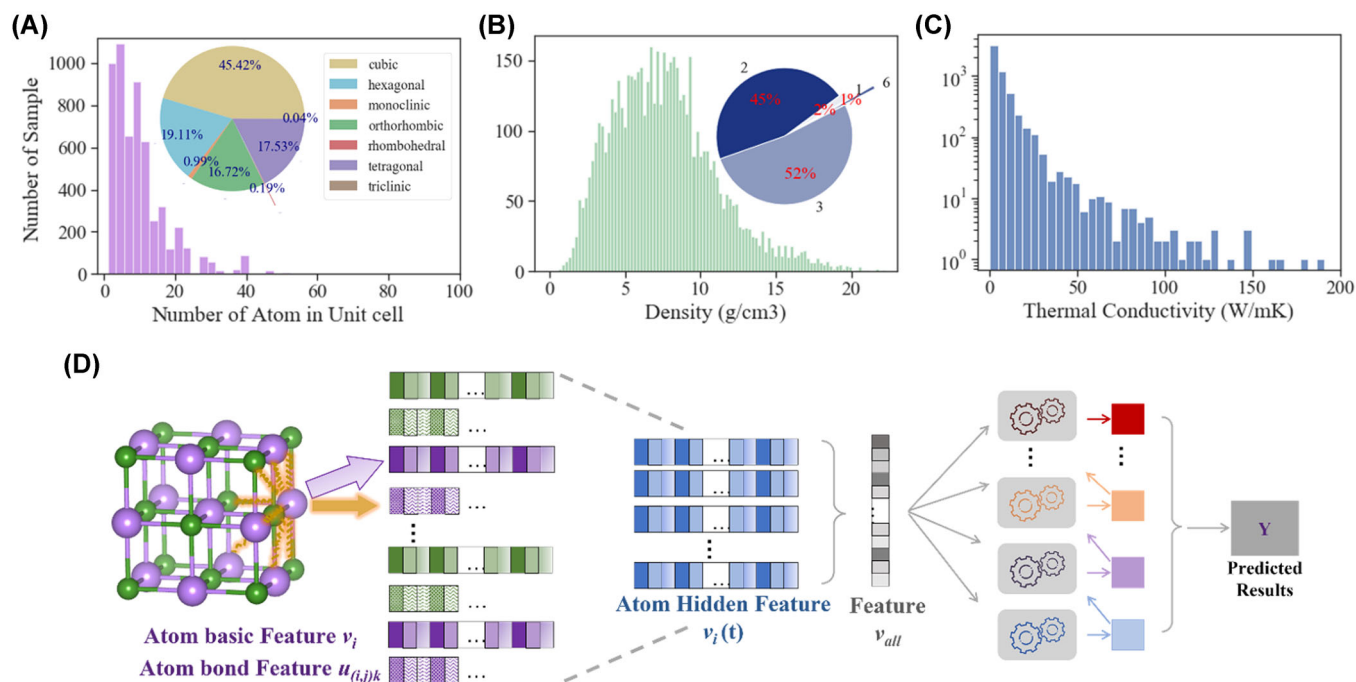


FIGURE 1 (A–C) Basic information of the lattice thermal conductivity data set. (A) The distribution of atomic number per cell and the insert pie chart give the sample occupancy of different crystal systems. (B) The density distribution and the element species of sample. (C) Thermal conductivity distribution. (D) The workflow of the machine learning algorithms framework

The structure of convolution network is composed of common convolution layers and pooling layers. Through convolution, the bonding hidden information could be encoded to each atom. With the pool layer, we could obtain the material eigenvector at a fixed dimension. The convolutional layers iteratively update the atom feature vector $\mathbf{v}_i$ by “convolution” with surrounding atoms and bonds with a nonlinear graph convolution function,

$$\mathbf{v}_i(t+1) = \text{Conv}[\mathbf{v}_i(t), \mathbf{v}_i(t), \mathbf{u}_{(i,j)_k}] \quad (3)$$

For each convolution process t , the atom feature vector $\mathbf{v}_i(t)$ can iteratively learn the information including its basic features and the hidden surrounding environment. The pooling layer is then used for producing the overall feature vector $\mathbf{v}_{\text{all}}$ for the crystal material, represented as,

$$\mathbf{v}_{\text{all}} = \text{Pool}(\mathbf{v}_0(0), \mathbf{v}_1(0), \dots, \mathbf{v}_N(0), \dots, \mathbf{v}_i(t), \mathbf{v}_N(T)) \quad (4)$$

Considering the insufficient samples in material data set, neural networks usually cannot be adequately trained. Only one simple learner is difficult to obtain the comprehensive and accurate results as well. We adopt an ensemble learning algorithm based on the boosting framework. Boosting algorithm series connect several weak learners, and train them to be a stronger one. Each

weak learner will adjust itself according to the previous results to reduce the overall bias.²⁷ Extreme gradient boosting algorithm was then introduced to obtain the target properties.

3 | RESULTS AND DISCUSSION

3.1 | Lattice thermal conductivity

The results of training the whole data set with the algorithms framework were shown in Figure 2A and Table 1. When the whole TC samples were used in training, evaluating, and testing, the coefficient of determination (R^2) of the training model can reach 0.999. At meanwhile, the mean absolute error (MAE) is only 0.39 W/(m K). Here, the scores of our model are better than previous results obtained by training with a similar data set.⁵ Their best results are 0.98 and 0.95 W/(m K), respectively. Then, the best-performing trained model was applied to predict the lattice TC of hundreds (565, 10% of the sample) of completely stranger materials. The mean error is only 2.747 W/(m K), indicating its good generalization ability.

Generally, evaluating and selecting training data with a standard criterion is beneficial to the performance of task-special models. To obtain a more professional training

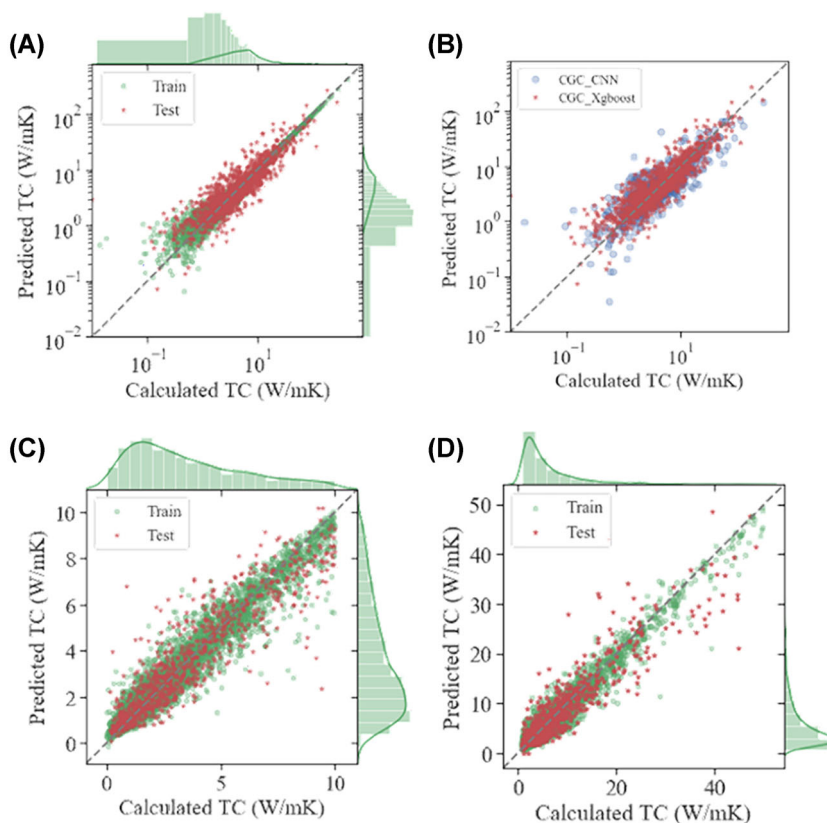


FIGURE 2 The training results of thermal conductivity predictions. (A, B) Results trained with all training samples. (C) Results trained on data set with low thermal conductivity (<10 W/(m K)) samples. (D) Results trained on data set with normal thermal conductivity (1–50 W/(m K)) samples

TABLE 1 The training and predicting results of thermal conductivity

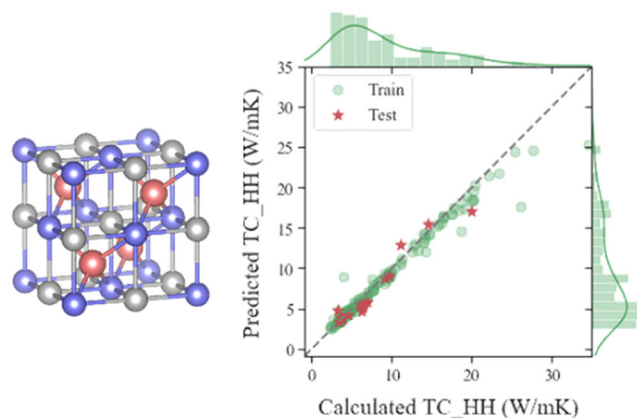
	Training		Testing	
	MAE	R^2	MAE	R^2
Whole set	0.393	0.999	2.747, (3.174) ^a	0.741
Low TC	0.506	0.918	0.704, (1.127) ^a	0.834
Normal TC	1.132	0.950	1.735, (2.667) ^a	0.852
Half-Heusler	0.575	0.953	0.799	0.947

^aThe predicted results of CGC_CNN.

model, we have divided the original data set with various restrictions. For clean energy conversion, excellent thermoelectric materials usually possess lower lattice TC. Therefore, we retrained on the data set of low lattice TC (TC < 10 W/(m K)) materials and obtain a TC prediction model more suitable for thermoelectric functional materials. The R^2 of this low-TC training model can reach 0.919, with a smaller error of 0.506 W/(m K). At the same time, predicting unknown low TC materials with this model can also get a smaller average error of about 0.704 W/(m K).

What's more, because the thermal conductivities of most crystalline materials is ranging from a few to tens of W/(m K). Materials with TC lower than 1 W/(m K) are considered as extreme low-TC materials. Oppositely, those whose TC of hundreds of W/(m K) are regarded as high-TC materials. To obtain a more generalized TC predictive model, we selected samples with thermal conductivities between 1 and 50 W/(m K). As shown in Figure 2D, we trained and tested 4951 materials with TC between 1 and 50 W/(m K). The MAE is only 1.132 W/(m K), and has a good correlation, the training R^2 reaching 0.950. This model also has good scores for predicting unfamiliar materials, with an average error of about 1.735 W/(m K).

To further verify the scope of our crystal graph embedded framework, we then trained a more material-targeted model with data from different sources.³² Half-Heusler materials as a kind of novel functional material are reported to have a great potential in the fields of thermoelectricity and spin transport, and so on.^{33–36} Several research focused on the thermal transport properties of this kind of materials, and there are some machine learning models used to predict the lattice TC of them as well.^{37,38} Training, evaluating and testing samples are collected from the former work with first-principles calculation.³² With our machine learning framework, an excellent TC prediction model for Half-Heusler materials was produced in Figure 3. The R^2 can reach 0.953, and the average error is only 0.575 W/(m K). This trained model also has good generalization ability with prediction MAE of only 0.799 W/(m K).

**FIGURE 3** The crystal structure of Half-Heusler materials and the training and testing results of 143 first-principles calculation samples

We also found that after embedded crystal graph features, some simple weak learners can also get better prediction performance. For example, Figure 4A shows the training model obtained by the KRR algorithm. The R^2 of the trained model can reach 0.934. Although the scores are worse than the results of ensemble boosting algorithm, it was slightly better than that obtained by neural network training model, marked in blue. Figure 4B gives the prediction results of embedded KRR. The above results indicated that embedding the material crystal graph information into the algorithms framework in the form of graph convolution neural network is a really effective strategy to construct the lattice TC prediction models.

3.2 | Other thermodynamic and electronic properties

In addition to training and predicting lattice TC, our algorithms framework has good effects in training other thermodynamic and electronic properties as well.

Heat capacity is an important physical property that describes the ability of absorbing or releasing heat, which has a wide range of applications in the fields of science and engineering. We conducted model training on the data set targeted as room temperature heat capacity, getting satisfactory training results. It was shown in Figure 5A that the obtained coefficient of determination R^2 is as high as 0.992, and the MAE is only 0.886 kJ/cell. Besides, the Grüneisen parameter represents the relationship between the vibration property and the volume. It reflects the anharmonicity of materials to a certain extent, playing an important role in estimating the TC. With the proposed algorithms framework, we also obtained good results in training the Grüneisen

FIGURE 4 The training and testing results of different crystal graph embedded model

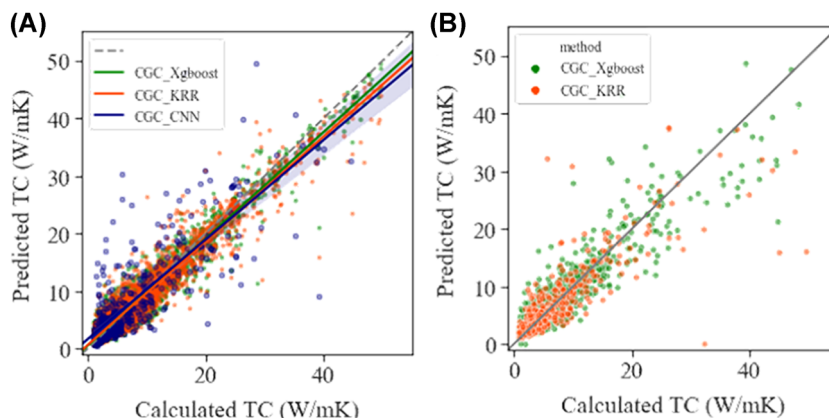
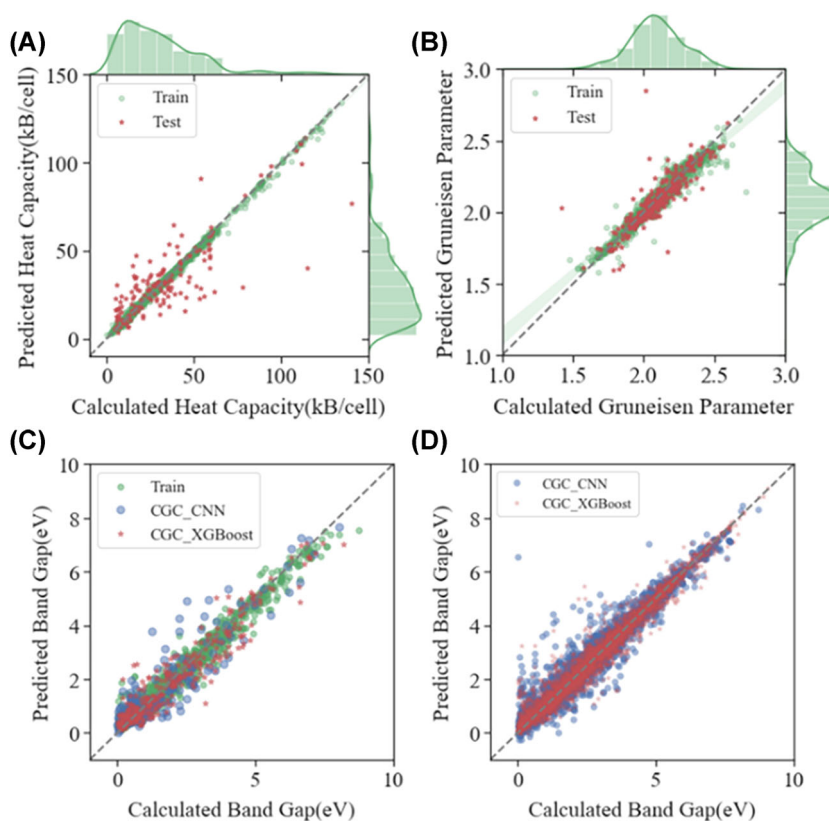


FIGURE 5 The training results of various phonon and electronic properties with our algorithm framework. (A) Heat capacity. (B) Grüneisen parameter. (C, D) Electronic bandgap



parameter. R^2 can reach 0.927 and the MAE is only 0.035, as shown in Figure 5B.

Furthermore, this algorithms framework was applied to train and predict the electronic bandgap of semiconductor materials. As shown in Figure 5C, the green dots represent the training data result, and the red points are the prediction result of the optimized model. Blues are the results of CGC_CNN training. In the selected data set, the R^2 of the training model is 0.990. And the average error of the tests is only 0.207 eV, which is better than 0.314 eV given by CGC_CNN model. These results indicated the good generalization ability and efficiency of our bandgap prediction model. We also collected more bandgaps samples, about 30,000 semiconductive samples from AFLOW, to enlarge our

training data set. More training samples can increase the coefficient of determination R^2 to 0.995. The prediction result of unknown data also shows a smaller MAE, 0.163 eV. These obtained surprising results not only verified the accuracy and generalization of our embedded algorithms framework, but also suggested their vast potential in ML model constructions of other phonon or electron correlation properties.

4 | CONCLUSION

In conclusion, we proposed a crystal graph embedded integrated algorithms framework to predict the TC of unrestricted solid materials. The framework only requires the

crystal structure information of the unknown crystallized material. Sufficient hidden information was extracted by the crystal graph convolution operation, and multiple single learners were integrated to enhance the learning ability of the whole algorithms framework. This machine learning framework was used to predict the TC of crystalized materials and present a good evaluation performance. The prediction coefficient of determination could reach 0.852, and the MAE was only 1.735 W/(m K) of materials with normal conductivities. Also, this trained model had a good generalization ability in unknown TC prediction. Besides, it was worth mentioning that the proposed machine learning framework can be used to predict not only the lattice TC but also other structure-related phonon and electronic properties, such as heat capacity, Grüneisen parameter, and electronic bandgap. This well-performance machine learning framework, combining the advantages of various algorithms, provides a novel approach for the application of machine learning in material science. We believe this study will inspire more excellent algorithms to promote the discovery of novel functional materials.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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